Do not write outside the box

0 4 This question is about the atmospheres of Earth and Mars.

0 4 . 1 Earth's early atmosphere may have been like the atmosphere of Mars today.

Why are scientists **not** certain about the percentage of gases in the Earth's early atmosphere?

[1 mark]

_____

_____

0 4 . 2 What was formed from the water vapour in the Earth's early atmosphere?

[1 mark]

Tick (✓) **one** box.

Crude oil ☐

Limestone ☐

Natural gas ☐

Oceans ☐

Do not write outside the box

0 4 . 3 The Earth's atmosphere today consists mainly of nitrogen and oxygen.

Draw **one** line from each gas to what produced the gas.

[2 marks]

Gas	**What produced the gas**
	Algae
Nitrogen	Animals
	Fossils
Oxygen	Oceans
	Volcanoes

Question 4 continues on the next page

Do not write outside the box

Table 2 shows the percentage of some gases in the atmospheres of Earth and Mars.

Table 2

Gas	Percentage of gas in atmosphere (%)	
	Earth	**Mars**
Argon	0.9	1.9
Carbon dioxide	0.04	95
Nitrogen	78	2.6
Oxygen	21	0.2

0 4 . 4 Why are animals **not** able to live on Mars?

[1 mark]

Tick (✓) **one** box.

The atmosphere of Mars does not contain enough argon. ☐

The atmosphere of Mars does not contain enough nitrogen. ☐

The atmosphere of Mars does not contain enough oxygen. ☐

0 4 . 5 There is more carbon dioxide on Mars than on Earth.

Which **other** gas is found in larger quantities on Mars than on Earth?

[1 mark]

_____

Do not write outside the box

0 4 . 6 Calculate how many times more nitrogen than oxygen there is in the atmosphere of Earth.

Use **Table 2**.

Give your answer to 2 significant figures.

[3 marks]

Number of times more nitrogen than oxygen (2 significant figures) = _____

9

Turn over for the next question